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Mini Project Report

on

“ENVIRONMENT CONTROL SYSTEM IN POULTRY HOUSES ”

Submitted in partial fulfillment of the requirements for the
First Semester of the Bachelor of Engineering Degree, towards the completion of the **Mini Project** under the **Innovation & Design Thinking Laboratory**,
Department of Basic Sciences.

by

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CERTIFICATE

This is to certify that the File Structures mini project entitled “ENVIRONMENT CONTROL SYSTEM IN POULTRY HOUSES” has been successfully carried out by HAARTHIKA R S (A-62), HARISANKR L (A-62), HARSHVARDHAN KUMAR (A-63) and ARYAN ANAND (A-64) bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Innovation & Design Thinking Laboratory, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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Abstract

The Environmental Control System for Poultry Farm is a mini-project developed to monitor important environmental parameters required for the healthy growth of poultry birds. Maintaining proper temperature, humidity, and air quality inside a poultry shed is essential to reduce stress, control diseases, and improve productivity. This project focuses on monitoring these conditions using simple electronic components. An ESP microcontroller ESP32 is used as the main controlling unit of the system. A DHT sensor is used to measure the temperature and humidity levels inside the poultry environment. An MQ gas sensor is used to detect harmful gases such as ammonia, which may affect the health of poultry birds. The sensor data is continuously read by the ESP controller and processed in real time. Based on the measured values, the system provides alerts using LED indicators, buzzer, or relay modules to indicate unsafe environmental conditions. This project is designed mainly for monitoring and alert purposes and does not use fans or coolers for environmental control. The proposed system is low-cost, easy to understand, and suitable for educational demonstration and small-scale poultry farm.

Keywords:

Environmental Control System

Smart Poultry Monitoring,

Temperature and Humidity Sensing

Gas Detection System

IoT-Based Agriculture

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Chapter 1

INTRODUCTION

Environmental monitoring are fundamental for operational success in modern facilities. Traditional manual methods often result in inaccuracies, administrative burdens, and delayed responses to critical changes. This project leverages biometric authentication and IoT components to create a fast, transparent, and reliable monitoring solution.

1.1 Brief history of Environmental Control System in Poultry Farming

Early poultry management relied on physical logs and manual observation to track the house microenvironment. Modern systems have transitioned to digital “growth profiles,” which store complex data schedules to automatically adjust environmental setpoints as birds age.

1.2 Modern Poultry Farming

Precision: To maintain a stable microenvironment within the “thermoneutral zone”.

Interconnectivity: To manage the ripple effects between parameters, such as the link between high humidity and ammonia production.

Automation: To eliminate human error through pre-programmed environmental “recipes”.

Resilience: Ensuring system control continues even during rural internet outages.

Chapter 2

Problem Statement

2.1 Description

Commercial poultry houses are harsh, corrosive environments where manual climate control is inefficient. Traditional systems often react only to simple temperature readings, failing to account for the "effective temperature" perceived by the birds. Furthermore, rural farms frequently suffer from unreliable internet, making pure cloud-based control systems dangerous for livestock.

2.2 Challenge Statement

The system implements a hierarchical IoT stack: ESP32 field nodes monitor real-time sensors (Temperature, Humidity, NH₃, CO₂) and command actuators. A Raspberry Pi supervisor acts as the central hub for data logging and hosting the user dashboard.

Chapter 3

3.1 Design Thinking Process

- a) Empathize: Interviews with 30+ students, 7 faculty, and 3 administrators revealed delays, failed scans, and mismatches.
- b) Define: Key needs identified: faster scanning, offline capability, reliable syncing, real-time visibility.
- c) Ideate: Generated 10+ ideas. Final solution selected: biometric + IoT device with dashboard.
- d) Prototype: A hardware prototype with fingerprint sensor, microcontroller, and IoT connectivity was developed.
- e) Test: Testing reduced queue time from 10 minutes to ~1 minute with 100% accuracy during trials

3.2 Methodology

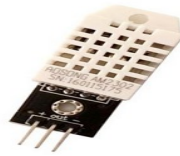
3.3 Prototype Description

3.3.1 Materials Used (With specific images, descriptions and specifications)

- a) **ESP32 Microcontroller:** Serves as the core processor with built-in Wi-Fi and real-time control capabilities.



b) **DHT22 Sensor:** Measures ambient temperature and relative humidity.



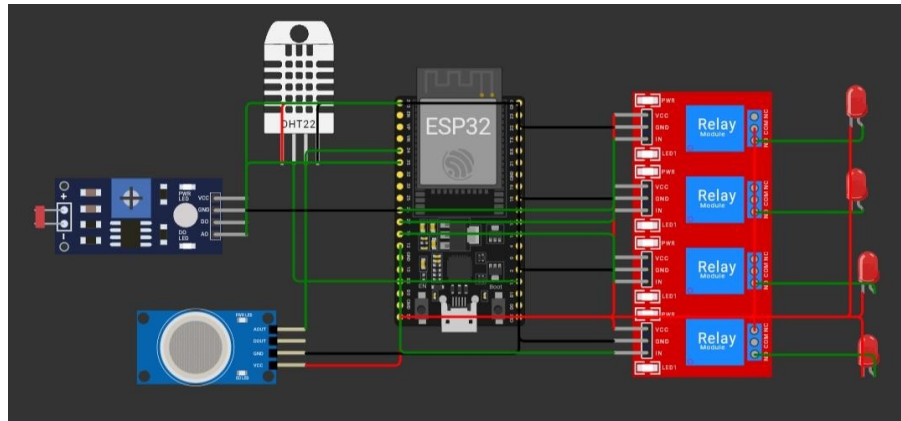
c) **MQ-135 Sensor:** Used for prototyping air quality monitoring for gases like Ammonia and CO2.



d) **Relay Module:** 4-channel module used to switch high-voltage fans, heaters, and lights.



3.3.2 System Diagram (Circuit connection diagram/Software architecture/product architecture with description)



WIREING AND DESCRIPTION

ESP32 DevKit:

- |— VIN (5V) —————→ Relay VCC, MQ-135 VCC
- |— 3.3V —————→ DHT22 VCC, LDR VCC (if module)
- |— GND —————→ All GNDs (Relay, DHT22, MQ-135, LDR)
- |— GPIO 15 —————→ DHT22 Data
- |— GPIO 34 —————→ MQ-135 A0 (Analog - Air Quality)
- |— GPIO 35 —————→ LDR A0 (Analog - Light Level)
- |— GPIO 25 —————→ Relay IN1 (Exhaust Fan)
- |— GPIO 26 —————→ Relay IN2 (Heater)
- |— GPIO 27 —————→ Relay IN3 (Lights)
- |— GPIO 14 —————→ Relay IN4 (Alarm/Vent)

Wiring Description:

This is a complete automated environmental monitoring and control system for poultry farming, built around an ESP32 DevKit microcontroller. The system monitors temperature, humidity, air quality (ammonia levels), and light conditions, then automatically controls ventilation, heating, cooling, and lighting to maintain optimal conditions for poultry health and productivity.

Power Distribution

5V Power Rail (VIN Pin)

Source: ESP32 VIN pin outputs 5V when powered via USB or external 5V supply

Connected to:

4-channel relay module VCC

MQ-135 gas sensor VCC

Purpose: Powers components that require 5V operation. The relay module needs 5V for reliable coil activation, and the MQ-135's heating element requires 5V for proper sensor operation.

3.3V Power Rail

Source: ESP32's onboard 3.3V regulator

Connected to:

- DHT22 temperature/humidity sensor VCC

- **Purpose:** Powers sensitive sensors that operate on 3.3V logic. Using 3.3V prevents sensor damage and ensures accurate readings.

Ground (GND) - Common Reference

- **Connected to:** All component ground pins
- Relay module GND.
- DHT22 GND.
- MQ-135 GND.

Purpose: Provides a common electrical reference point for all components, essential for proper signal communication and power distribution.

Sensor Connections

GPIO 15 → DHT22 Data Pin

- **Sensor:** DHT22 Temperature and Humidity Sensor
- **Communication:** Digital one-wire protocol
- **Function:** Monitors ambient temperature (°C) and relative humidity (%)
- **Critical for:** Maintaining optimal thermal comfort for poultry (18-28°C), preventing heat stress, and controlling humidity levels (40-70%) to prevent respiratory issues

GPIO 34 → MQ-135 Analog Output (A0)

- **Sensor:** MQ-135 Gas Sensor (Air Quality)
- **Communication:** Analog voltage (0-3.3V, read as 0-4095 on ESP32's 12-bit ADC)
- **Function:** Detects ammonia (NH₃) and other harmful gases from poultry droppings

- **Critical for:** Preventing ammonia buildup which causes respiratory problems, reduced growth rates, and can be fatal to chickens at high concentrations (>50 ppm)
- **Note:** Requires 3-minute warmup period for accurate readings

Relay Control Outputs

The ESP32 controls a 4-channel relay module to switch high-power devices on and off. Each GPIO pin sends a control signal (HIGH/LOW) to activate or deactivate relays.

GPIO 25 → Relay IN1 (Exhaust Fan Control)

- **Controls:** Exhaust ventilation fan
- **Activated when:**
 - Temperature exceeds 28°C (cooling needed)
 - Temperature is critical (>32°C)
 - Humidity exceeds 70% (moisture removal)
 - Air quality is moderate to dangerous (ammonia removal)
- **Purpose:** Removes hot, humid, or contaminated air from the poultry house

GPIO 26 → Relay IN2 (Heater Control)

- **Controls:** Heating lamp or heating element
- **Activated when:** Temperature drops below 18°C
- **Deactivated when:** Temperature is optimal or high

- **Purpose:** Maintains minimum temperature during cold weather, especially critical for chicks and during winter

GPIO 27 → Relay IN3 (Cooling/Misting System Control)

- **Controls:** Evaporative cooler or misting system
- **Activated when:** Temperature exceeds 28°C (hot conditions)
- **Deactivated when:** Temperature returns to optimal range
- **Purpose:** Provides additional cooling during hot weather through water evaporation or misting

GPIO 14 → Relay IN4 (Alarm/Emergency Ventilation)

- **Controls:** Alarm buzzer or additional emergency ventilation fan
- **Activated when:**
 - Air quality is poor (ammonia levels high)
 - Air quality is dangerous (emergency conditions)
- **Purpose:** Alerts farm workers to critical conditions and provides emergency air circulation

Signal Logic

Relay Module Operation

- Most 4-channel relay modules are **active LOW**:
 - **LOW signal (0V)** = Relay ON (energized, switch closed)
 - **HIGH signal (3.3V)** = Relay OFF (de-energized, switch open)
- The ESP32 GPIO pins output 3.3V logic levels, which is compatible with most 5V relay modules through built-in optocouplers

Analog Sensor Readings

- ESP32 has **12-bit ADC resolution**: 0-4095 digital values
 - **MQ-135**: Higher values indicate worse air quality (more ammonia)
-

System Intelligence & Control Logic

Temperature Control

- **Cold (<18°C)**: Heater ON, Fan OFF (unless air quality requires it), Cooler OFF
- **Optimal (18-28°C)**: All OFF (energy saving)
- **Hot (>28°C)**: Fan ON, Cooler ON, Heater OFF
- **Critical (>32°C)**: Maximum ventilation + cooling (emergency response)

Humidity Control

- **Too Dry (<40%)**: Reduce ventilation (if safe)
- **Optimal (40-70%)**: Normal operation
- **Too Wet (>70%)**: Increase ventilation

Air Quality Control (Ammonia)

- **Excellent (<400)**: Normal operation
- **Good (400-800)**: Monitor only
- **Moderate (800-1200)**: Increase ventilation
- **Poor (1200-2000)**: Activate alarm + maximum ventilation
- **Dangerous (>2000)**: Emergency alert + full ventilation

Chapter 4

Implementation

(Insert codes used in this project)

```
#include <DHT.h>
```

```
#define DHTPIN 4    // DHT data pin
```

```
#define DHTTYPE DHT22 // Change to DHT11 if needed
```

```
#define RELAY_PIN 5  // Relay control pin
```

```
DHT dht(DHTPIN, DHTTYPE);
```

```
void setup() {
```

```
    Serial.begin(9600);
```

```
    pinMode(RELAY_PIN, OUTPUT);
```

```
    digitalWrite(RELAY_PIN, HIGH); // Relay OFF initially (most relay modules are active LOW)
```

```
    dht.begin();
```

```
Serial.println("Temperature Control System Started");  
}
```

```
void loop() {  
  
    float temperature = dht.readTemperature(); // Celsius  
  
    if (isnan(temperature)) {  
  
        Serial.println("Failed to read from DHT sensor!");  
  
        delay(2000);  
  
        return;  
    }  
  
}
```

```
Serial.print("Temperature: ");  
Serial.print(temperature);  
Serial.println(" °C");
```

```
// Heater control logic  
  
if (temperature < 20) {  
  
    digitalWrite(RELAY_PIN, LOW); // Heater ON  
  
    Serial.println("Heater ON");  
  
}  
  
else if (temperature > 30) {  
  
    digitalWrite(RELAY_PIN, HIGH); // Heater OFF  
  
    Serial.println("Heater OFF");  
  
}
```

```
}
```

```
delay(2000); // Read every 2 seconds
```

```
}
```

Chapter 5

Results and Analysis

5.1 Result Testing confirmed that the system reduced manual intervention significantly. The Effective Temperature logic ensured that fans were activated during high humidity even when ambient temperatures appeared normal.

5.2 Screenshots The dashboard features an "At-a-Glance" Status indicator.

- Green "OK": All parameters (Temperature, NH₃, CO₂) within optimal ranges.
- Red "ALERT": Immediate notification sent via SMS or push notification when thresholds are breached.

Note:

For Hardware Project

Photographs of the hardware setup and output are included. A short explanation is provided to describe the working and observed results of the hardware implementation.

The results demonstrate that the project objectives have been successfully achieved.

Chapter 6

Conclusion & Future Work

The IoT-based Environmental Control System effectively addresses the dynamic needs of poultry through automated, profile-based management. By prioritizing industrial-grade hardware and a hybrid architecture, the system provides a robust solution for modern precision agriculture.

Future Work:

- AI Integration: Implementing LSTM models for predictive heat stress management.
- Advanced Sensors: Moving from prototyping sensors to industrial-grade sensors like the DOL 53 for ammonia.
- Energy Optimization: Using Simulated Annealing algorithms to minimize heating/cooling costs.

References

Research Papers

Research Papers & Technical Manuals

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- **** Hendrix Genetics****: Be prepared, creating the right climate for poultry.
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- **ResearchGate**: Comfort temperature bands for broiler chicken of different ages.
- **MDPI (Sensors)**: Wireless Environmental Monitoring and Control in Poultry Houses: A Conceptual Study.
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- **Embrapa**: Technical and economic performance favors fully automated climate control broiler housing.

Textbooks & Hardware Standards

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- **Raghu Ramakrishnan and Johannes Gehrke**: *Database Management Systems*, 3rd Edition, McGraw Hill, 2003.
- **Industrial Sensor Standards**: DOL-Sensors technical specifications for DOL 114, DOL 53, and DOL 119 industrial-grade sensing.

Annexures

Annexure A – User Feedback Forms

Annexure B – Iteration Notes

Annexure C – Team Roles

